QSAR-Based Prediction of Alzheimer's Disease Bioactivity Using Morgan Fingerprints and Molecular Descriptors with Machine Learning"

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ABSTRACT

Alzheimer's disease (AD) is a progressive neurodegenerative condition that leads to loss of memory and deteriorating cognitive ability. Even though , till date there is no proper treatment for this disease. Acetylcholinesterase (AChE) remains as one of the primary targets for therapy, especially in the symptomatic mitigation. The aim of this study is to create an insilico pipeline to predict the bioactivity of the molecules that work against AchE target.

Ligands were sourced from the BindingDB database and carefully preprocessed by filtering for duplicates, removing invalid structures, and converting IC_{50} values to pIC_{50}, yielding a final dataset of 6,641 compounds. For the feature extraction process, three different paths were examined using RDKit and PaDELPy: Morgan Fingerprints (radius = 2, 2048-bit vectors), about 208 features were generated, RDKit molecular descriptors, about 185 features were generated, and molecular descriptors from PaDELPy resulted in 1878 features Seven machine learning algorithms were trained with an 80:20 train/test split and evaluated using fivefold cross-validation. These include Linear Regression, Ridge, Lasso, K-Nearest Neighbors (KNN), Support Vector Regression (SVR), Random Forests, and Gradient Boosting. Among all these,the Random Forest model using Morgan fingerprints achieved the highest performance (R^2=0.7283, RMSE=0.8041). The next performing model was the Extra Trees model ith PaDELPy descriptors (R^2 = 0.6942, RMSE = 0.8001), and Support Vector Regression performed best with RDKit descriptors (R^2 = 0.6526, RMSE = 0.8951) In conclusion, fingerprint representation proved more efficient than descriptor-based approaches, suggesting that structural properties effectively capture the structure-activity relationship of acetylcholinesterase inhibition. These findings highlight the importance of machine learning-based QSAR models as an effective means of bioactivity prediction and virtual screening in the context of Alzheimer’s disease drug discovery Keywords: Alzheimer’s disease, Acetylcholinesterase, QSAR, Morgan, Fingerprints, RDKit, PaDELPy, Machine Learning, Random Forest, Extra Trees, Computer-Aided Drug Design (CADD)

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INTRODUCTION

1.1 ALZHEIMER'S DISEASE: A GLOBAL HEALTH CHALLENGE

Among the neurodegenerative disorders that exist, Alzheimer's disease (AD) stands out the most as the most common leading cause of dementia , which affects almost 60-70% of elderly individuals [1]. This develops with minor memory lapses which slowly worsens with the time that, eventually leads to severe cognitive issues like memory loss , changes in the way of talking and behaviour affecting the daily functioning of the individuals suffering from this. Since there isa decline in the mortality rate the life expectancy has risen due to this AD has become a major socioeconomic problem in the healthcare system. As of now, the current treatments for Alzheimer's disease only manage the symptoms; the disease itself is not addressed [2].

1.2 ACETYLCHOLINESTERASE (AChE) AS A THERAPEUTIC TARGET

1.2.1 The Cholinergic Hypothesis

The idea behind the current treatment of AD is the cholinergic hypothesis. This hypothesis states that the cognitive decline in people with Alzheimer's disease is mostly caused by the slow loss of cholinergic neurons in the basal forebrain, which results in the reduction of the acetylcholine (Ach) levels in the brain [3]. Acetylcholinesterase (AchE) is the primary enzyme that catalyses the breaking down acetylcholine into choline and acetate in the synaptic cleft , thus stopping the cholinergic neurotransmission. This process is very much essential for the normal brain functions like memory, learning, and attention span. In AD, the loss of the cholinergic neurons and the increased activity of acetylcholinesterase reduces the acetylcholine levels even more; this reduction evokes the symptoms which the AD patients usually experience[4].

1.2.2 The Structural Basis: Dual Binding Sites of AChE

The unique structure of AChE makes it a good target for the drug development/manipulation. Enzyme has a deep, narrow tunnel, i.e., 20 units long, it runs from the surface of the enzyme to its catalytic center, where the reactions take place, and is lined with many aromatic acid residues, two primary binding sites are identified within this gorge as mentioned below:

Table 1.1 *Binding regions of acetylcholinesterase and their respective amino acid compositions and function*

Binding Site	Location	Key amino acids	Primary function
Catalytic Active Site (CAS)	Bottom of the gorge	S203,H447,E334,W86	Hydrolyze the acetylcholine
Peripheral Anionic Site (PAS)	Entrance of the gorge	Y72,Y124,W286,Y341	Promotes the Aβ aggregation

The peripheral anionic site (PAS), particularly residues such as W286 and Y341, plays a role in ligand binding by facilitating π–π interactions with aromatic ligands. These interactions make sure that the ligand is in the position towards the catalytic active sites. The binding of PAS and Cas leads to the dual-binding structure in Ache, which makes it a excellent fit for the inhibitor development and opens up novel options for the drug design, as dual site inhibitors binding to both sites at the same time [5]. Such Structural features makes it suitable for the QSAR modeling because of the relevance of the molecular descriptors related to the aromaticity, distribution of electrons, and configuration , which is directly linked to the interaction between ligand binding at PAS and CAS region, which affects the accuracy of the IC_{50}/pIC_{50} values.

1.2.3 The Dual Role of AChE

Recent research revealed that the AChE plays a major role in the disease in Alzheimer's disease pathology, besides its classical catalytic function. The peripheral anionic site (PAS) of AChE can bind β-amyloid (Aβ) peptides, allowing them to cluster and form neurotoxic amyloid plaques, a key feature of the disease. Moreover, AchE has been observed to exhibit chaperone-like properties, in which the enzyme interacts with amyloid aggregates and promotes fibril formation. AchE also contributes directly to Aβ aggregation via

PAS-mediated interactions, thereby highlighting its relevance as a multifaceted therapeutic target for Alzheimer's disease [6].

This observation s further strengthens the designation of AchE as a dual target:

- Cholinergic deficiency through CAS: leads to cognitive decline
- Amyloid plaque formation through PAS leads to the disease progression [4]

1.2.4 Current AChE inhibitors and their limitations:

Based on the cholinergic hypothesis, currently four acetylcholinesterase inhibitors have been approved for the clinical use [4].

Table 1.2 *Clinically approved AChE inhibitors for Alzheimer's disease:*

Drug	Approved Year	Mechanism	Selectivity	Key limitation
Tacrine	1993	Reversible AchE inhibitor	Non-selective	Withdrew from the market due to hepatotoxicity
Donepezil	1996	Reversible AchE inhibitor	About 1000 times selective for AchE	Gastrointestinal side toxicity
Rivastigmine	1997	Pseudo-irreversible AchE inhibitor	AchE /BuChE inhibition	Frequent gastro-intestinal toxicity
Galantamine	2001	Reversible AchE inhibitor and nicotinic receptor modulator	AchE-selective	Gastrointestinal side effects

However, despite their clinical applications, these drugs have major limitations. First, they provide symptomatic relief by improving the cholinergic neurotransmission, but fail to tackle the neurodegeneration and the disease progression [1][2]. Moreover, most inhibitors used today act primarily on the CAS and do not affect the PAS, which is involved in β-amyloid aggregation [5]. Usage of these drugs can cause side effects like nausea, diarrhea, and vomiting due to the excessive systemic activation of cholinergic transmission [2]. The first drug, Tacrine, was withdrawn from clinical use due to its hepatotoxicity [7].

Overall, the current therapeutic strategies, such as acetylcholinesterase inhibitors and N-methyl-D-aspartate receptor antagonists, are mainly symptomatic and unable to modify Alzheimer's disease progression [2].

1.2.5 Why AChE Remains an Attractive Target for the Multi-Target-Directed Ligand Development.

Despite all the limitations of current treatments, acetylcholinesterase (AChE) remains a highly promising therapeutic target in AD. Even though the current drugs remain imperfect, they provide reliable results in improving cognitive ability. In addition, AchE plays an important role in the two major mechanisms of the pathology: cholinergic deficiency via CAS and amyloid aggregation via PAS. The specific dual-site structure of AchE can serve as a basis for developing multi-target-directed ligands that interact with the CAS and PAS simultaneously, offering a significant advantage in treating complex diseases, with enhanced therapeutic efficacy and reduced side effects [1][2][5].

1.3: CADD and Machine Learning in Drug Discovery

1.3.1 Computer-Aided Drug Design (CADD)

Conventional drug discovery is a long and expensive process that takes about 10-15 years for completion and costs over $2 billion to just bring out one drug into the market [8.]. This

problem is more common in neurological disorders like Alzheimer's disease, which is often connected with the high failure rates in clinical trials because of its complexity [1].

To solve this challenge, one of the most efficient techniques is Computer-Aided Drug Discovery. CADD mainly involves in using different in silico approaches that helps in accelerating drug discovery, thereby reducing the cost and time [8].These techniques facilitate quick virtual screening of vast libraries of chemicals for the identification of promising candidates before the experimental validation [9]. It is important to note that such computational-based approaches play a key role in ligand-based modeling, and are the foundation of QSAR.

1.3.2 Machine Learning Models for Bioactivity Prediction

In the broader CADD context, Machine Learning (ML) has become increasingly relevant, especially in QSAR modeling [10]. The process involves training machine learning algorithms on available databases that show a correlation between structural information and biological activity to predict biological activity and provide results such as PIC50, thereby greatly reducing the number of compounds required for experimental evaluation [11].

Several studies have been published highlighting the applicability of M models for predicting inhibitors of acetylcholinesterase (AchE) and related targets. For instance, Hou et al. (2025) used ML algorithms such as Random Forest and Decision Tree for predicting dual AChE/MAO-B inhibition [12]. Bozkır et al. (2025) used LightGBM along with the molecular fingerprints to identify dual AChE/BACE1 inhibitors[13]. Similarly, Maliyakkal et al. (2025) used an artificial neural network model to predict GSK-3β inhibitors [11]. Thus, overall, we can say that the ensemble-based methods frequently achieved strong predictive performance in bioactivity modeling tasks [11][12]

Based on these insights, this project employed almost seven models across three different methodologies to facilitate the comparative analysis of predictive performance in modeling AchE inhibition:

Category	Models Included	Purpose in context with this study

Linear models	Linear Regression, Ridge, Lasso	Provide a baseline for the predictive capabilities
Non-Linear models	Support Vector Regression (SVR), K-Nearest Neighbors (KNN)	Capture non-linear structure-activity relations
Ensemble methods	Gradient Boosting, Random Forest, Extra Trees	Better predictive accuracy and robustness

Table 1.3: Categories of predictive models used in this study.

This justifies the use of ML-QSAR models for predicting the inhibition of AchE in the current study

1.3.3 Feature extraction methods

Since machine learning algorithms operate only on numbers, chemical structures must be represented as numerical features via feature extraction [14]. There are several proven methods to do this in cheminformatics, namely the RDKit descriptors, PaDEL descriptors, ECFP (Extended-Connectivity Fingerprints)[15][16][17]

The representations mainly include essential structures and physicochemical properties such as substructure motifs, electronic characteristics, polarity, and molecular size. Previous studies have found that substructure-based fingerprints are highly effective for predicting IC50 values, as they capture pharmacophore-relevant features that are closely involved in target interactions [18]. In this project, these feature extraction techniques are applied to convert SMILES representations into numerical descriptors suitable for developing the QSAR model.

1.3.4 QSAR and Its Role in AChE Inhibitor Discovery

The Quantitative Structure–Activity Relationship (QSAR) modeling approach is one of the most fundamental ligand-based methods in CADD, aiming to establish quantitative relationships between molecular structure and biological activity [14]. The analysis of

molecular descriptors and their correlation with the measured activity allows us to predict PIC50 values and facilitate rational drug design [14][10].

For AchE inhibition, ML-driven QSAR models provide an efficient way to screen large compound datasets. They help in identifying the important molecular descriptors as well as support the selection of potential lead molecules for further analysis .Thus, QSAR is the primary strategy in this study for the prediction of IC_{50}/pIC_{50} values of the potential AchE inhibitors

1.3.5 Why Machine Learning for AChE Inhibitors?

Alzheimer's disease involves multifactorial pathology, whereby a therapy targeting one component might not be enough to handle the disease progression [9][1]. The need for treatment strategies targeting multiple targets, along with the computational methods, is increasing.

One such approach is the design of multi-target-directed ligands (MTDLs), which aim to simultaneously target multiple disease targets. This approach would not only increase treatment effectiveness but also minimize side effects [1]. Computational methods such as QSAR, molecular docking, and virtual screening have played important roles in the discovery of such ligands [9]

In addition, recent work has highlighted the application of machine learning models to predict dual-target inhibition. This shows the promising aspects of machine learning models in discovering new drugs

Hence, this study aims at predicting the activity of the compounds that inhibit AchE via the machine learning-based QSAR modeling using molecular descriptors obtained from SMILES

1.3AIMS AND OBJECTIVES

AIM

The aim of this project is to develop and evaluate machine-learning-based QSAR models for predicting IC50/pIC50 of the compounds that target acetylcholinesterase(AchE) based on molecular descriptors calculated from the SMILES representations

OBJECTIVES

1. To collect the dataset of AchE inhibitors containing experimentally IC50 from the Binding DB database
2. Preprocess the dataset that included steps like cleaning the SMILES structure, eliminating the duplicates, removal of missing data points nd conversion of IC_{50} values to pIC_{50} values.
3. Generating molecular descriptors and fingerprints from SMILES strings using cheminformatics packages. Morgan fingerprints generated by RDKit, RDKit molecular descriptors, and PADELPy molecular descriptors
4. Implementation and comparison of multiple machine learning algorithms, including linear (Linear Regression, Ridge, Lasso), non-linear (KNN, SVR), and ensemble methods (Random Forest, Gradient Boosting).
5. To assess the performance of models using R^2, RMSE, and MAE metrics
6. For selecting the best model for the prediction of AchE inhibitory activity by comparing the performances of the models
7. To analyze the relationship between molecular features and the inhibitory activity in order to gain deeper insight into structure-related activity
8. To predict the IC_{50}/pIC_{50} value of unknown compounds from their SMILES representations using the model we have developed

2. MATERIALS AND METHODS

2.1 Dataset Acquisition

The dataset utilized in this project was downloaded from the Binding DB. (https://www.bindingdb.org), which is an open-access database of experimentally determined binding affinity values of the protein-ligand interactions. The dataset (Alzheimer's dataset.tsv) holds interaction values of compounds screened against the acetylcholinesterase enzyme (AchE), a well-established target in the disease treatment. The original data set had about 9,388 rows and 640 columns containing ligand, SMILES, and IC50 values and other metadata.

2.2 Data Preprocessing and Curation

2.2.1 Column Selection

To make the prediction of how well the molecule can inhibit, the details mentioned below were picked from the dataset :

- Ligand SMILES- canonical molecular representation
- IC_{50} (nM) — a quantitative measure for the inhibitory activity
- BindingDB Reactant Set ID — a unique identifier assigned to each compound

2.2.2 Handling Missing and Duplicate Entries

Missing data was deleted using the dropna() function making the data complete. Futhermore, data with the same SMILES string were removed to reduce the possible bias during the training process as well to prevent the data leakage.

2.2.3 Cleaning of IC_{50} Values

The IC_{50} values were initially presented as a string, which also included the inequality operators (>,<,~), which were removed from the making it more cleaner, then converted to a numeric data type by calling the function pd.to_numeric(values. errors='coerce') on each of the values. The values that did not convert were totally removed.

2.2.4 Cleaning of IC_{50} Values

In order to overcome the greater level of variation among the IC_{50} values, this transformation was done with the help of the logarithm to achieve to pIC values

$$pIC_{50} = \; - \left(IC_{50} \frac{(nM)}{10^9} \right)$$

This transformation gave a continuous scale, higher the pIC50 values indicating as a more potent inhibitory . After preprocessing of the data, our final dataset conists of about 6,641 compounds with SMILES notation and corresponding to the pIC50 values.

2.3 **Molecular Representation**

SMILES strings were parsed into RDKit molecular structures using Chem.MolFromSmiles(). The entries with invalid SMILES were removed from the dataset, resulting in 6,640 valid molecular structures.

2.4 **Feature Extraction**

To fully represent molecular structure, three feature-extraction methods were used, each highlighting different aspects of chemical information.

2.4.1 Morgan fingerprints

The Morgan fingerprints (radius=2, 2048-bit) were generated using RDKit. These fingerprints provide a binary representation of the local atomic environment, which helps represent substructural patterns. Features with zero variance were excluded from the dataset, resulting in a final set of 2,047 informative features.

2.4.2 RDKit Molecular Descriptors

The RDKit library was used to compute 217 molecular descriptors of each compound. These descriptors are different physicochemical and structural properties of molecules. Before modeling, a number of preprocessing steps were used to guarantee the data quality and reliability. NaN was first introduced to replace the infinite values in order to avoid computational errors. The descriptors whose percentage of missing values exceeded 10

percent were then filtered out to keep the descriptors with a high enough percentage of non-missing values. To narrow down the feature set further, low-variance filtering has been used with a threshold of 0.01. The step removed descriptors that vary little among the data, as they do not give any meaningful information to help predict. The final RDKit descriptor set had 185 features, and they were used to further develop the model

2.4.3 PaDEL Descriptors

The molecular descriptors were generated through the padelpy Python interface using the PaDEL-Descriptor software. To achieve stability during computations and avoid memory limitations in the Google Colab environment, the SMILES were split into batches of 200 molecules. In total, 36 batches were generated. A custom batch-processing method was implemented, and intermediate checkpoint CSV files were saved after processing every 50 molecules, which were later combined into a single consolidated dataset.

The initial set contained 1,878 molecular descriptors per compound. The dataset, after combining the output from all the batches, went through the following preprocessing :

Numeric conversion: All the descriptors' columns were converted to numeric values, and all the non-numeric values were converted to NaN

Infinite value Handling: Inf values (np.inf, -np.inf) were replaced by NaN.

Missing value filtering: The descriptors with more than 10% missing values were dropped

Removal of Zero-variance: Descriptors that had constant values across the samples were eliminated, resulting in removal of 213 features

2.5 **Data Splitting**

Each data set from the feature sets was split into 80% for training and 20% for testing using the train_test_split method, with a random state of 42.

2.6 **Model Development and Training**

2.6.1 Models Implemented

Eight models have been used for regression, which include different learning methods:

Table 2.1 Categories of machine learning models used in this study

Category	Models
Linear	Linear Regression, Ridge and Lasso
Distance	K-Nearer neighbors (KNN)
Kernel	Support Vector Regression (SVR) using RBF kernel
Ensemble	Random Forest and Extra Trees (both bagging algorithms) and Gradient Boosting (a boosting algorithm)

2.6.2 Pipeline Architecture

The implementation of each model consists of utilizing a Pipeline from scikit-learn, which consists mainly of two stages:

- StandardScaler: used for normalizing the feature set to have a mean of 0 and variance 1
- Regression model : the predictive model

2.6.3 Training and Cross-validation

The model's performance was analyzed using a 5-fold cross-validation (KFold, shuffle=True, random_state=42), with R^2 as the metric. The models were first trained on the training set, then tested on the test set.

2.7 **Model Evaluation Metrics**

The performance of the models was assessed based on two criteria that capture both the prediction error and explanatory power.

R^2 (Coefficient of Determination)

$$R^2 = 1 - \frac{\Sigma(y_i - \hat{y})^2}{\Sigma(y_i - \bar{y})^2}$$

Higher values of R^2 (with the maximum value of 1) mean that the variance explained by the model is high, indicating good predictive ability

RMSE (Root Mean Square Error)

$$RMSE = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2}$$

The lower the RMSE, the smaller the prediction error, as measured by the pIC_{50}.

Along with the above-mentioned measures, the results of cross-validation were presented as the mean R^2 and its standard deviation to assess the model stability and generalization.

2.8 **Software and Computational Environment**

The analysis was conducted using Python 3.10 within the Jupyter Notebook environment , with the use of several specialized packages that supported the different stages of the pipeline:

Table 2.2 Software libraries used in the machine learning workflow and their purposes.

Library	Purpose
RDKit	Feature generation, molecule parsing, Morgan fingerprints
PaDEL-Descriptor (via PaDELpy)	Molecule descriptor calculation
Pandas, NumPy	Data handling and numerical computations
Scikit-learn	Building models, feature scaling, and model evaluation
Matplotlib, Seaborn	Data visualization

The workflow was structured to be fully reproducible. During the descriptors generation using PaDEL , batch processing and checkpoints were employed to enhance the computational reliability and reduce the possibility of interruptions or the loss of data

3. Results and Discussion

3.1 Dataset Overview

The initial dataset, extracted from BindingDB, contained about 9,388 compounds, each with an IC_{50} value and a molecular structure encoded as a SMILES string. Multiple pre-processing steps were applied, including the removal of missing values, duplicates, and invalid IC_{50} values, resulting in about 6,641 compounds with valid pIC_{50} values. In order to enable regression modeling, the IC_{50} values were transformed into pIC_{50} values by employing a log transformation. This helps in avoiding any skewness in the distribution if it exists and obtaining normal data, which will enhance the model quality and predictability. The histogram of pIC_{50} values from the processed dataset is shown in Figure 3.1

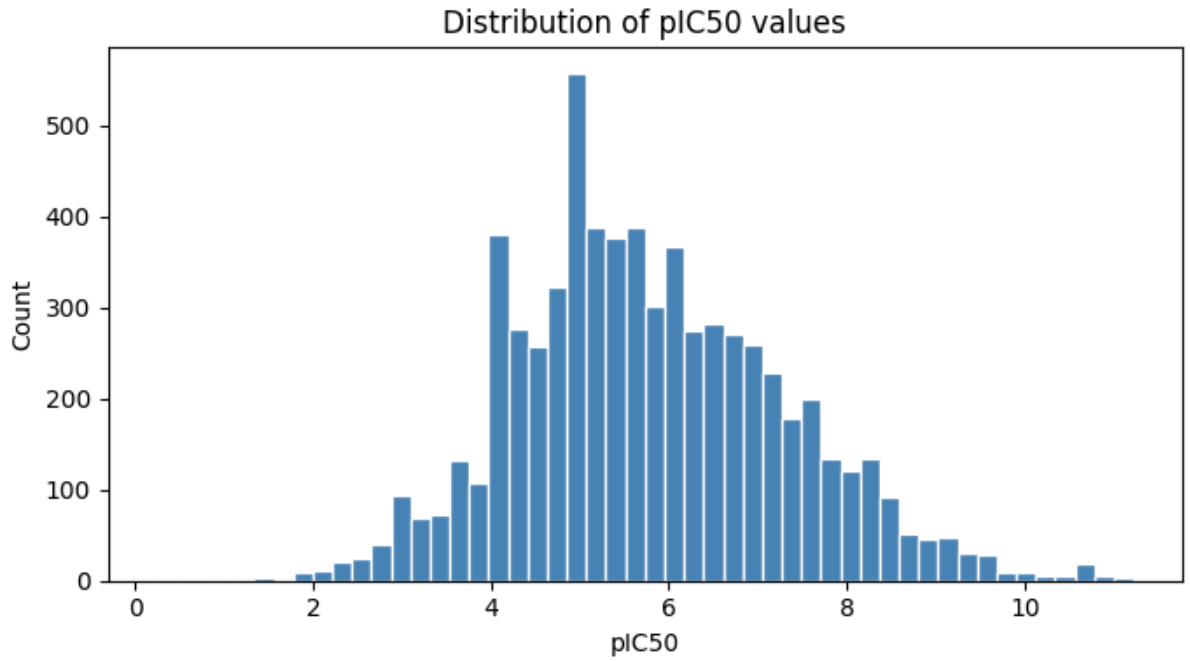


Figure 3.1: Histogram of pIC_{50} values in the dataset.

The distribution of values in the dataset is slightly bell-shaped, indicating that the majority of molecules have medium activity levels, making the dataset suitable for use in ML-based QSAR modeling

3.3 Model Performance using Morgan Fingerprints

3.2.1 Feature Overview of Morgan Fingerprints

The circular fingerprints (also referred to as Morgan fingerprints) were employed to depict molecular structures in a numerical form that could be analyzed by a machine learning model. These fingerprints encode local structural information, where the presence or absence of substructures around each atom in a given radius is coded. In this test the RDKit library was used with a radius of 2 and a constant length of 2048 bits to generate Morgan fingerprints. The bits in a fingerprint vector represent the presence of a particular chemical substructure where a 1 indicates the presence of that substructure and a 0 indicates the absence of that substructure. To enhance the quality of features, zero-variance features (bits that do not change based on the different molecules) were eliminated, because they do not affect learning of the model. The end result of this filtering step was a final feature set of 2047 informative features. These fingerprints allow a compact and efficient representation of the topology of the molecule, allowing machine learning models to effectively capture structure-activity relationships. They are especially good at QSAR modeling tasks because of their capabilities to encode local chemical environments.

3.3.2 Model Performance Results

The performance of the models using Morgan fingerprints is presented in table 3.1 below

Table 3.1: Model performance using Morgan fingerprints

Model	CV R^2 Mean	CV R^2 Std	Test R^2	Test RMSE
Random Forest	0.7069	0.0170	0.7283	0.8041
SVR	0.6753	0.0123	0.7033	0.8403
Lasso Regression	0.6130	0.0235	0.6370	0.9295
KNN Regressor	0.5719	0.0221	0.6361	0.9306

Gradient Boosting	0.5096	0.0297	0.5367	1.0500
Ridge Regression	-0.0815	0.1667	0.3636	1.2307
Linear Regression	-0.1300	0.1827	0.3558	1.2381

The relative performance of the various models is shown in Figure 3.2 below.

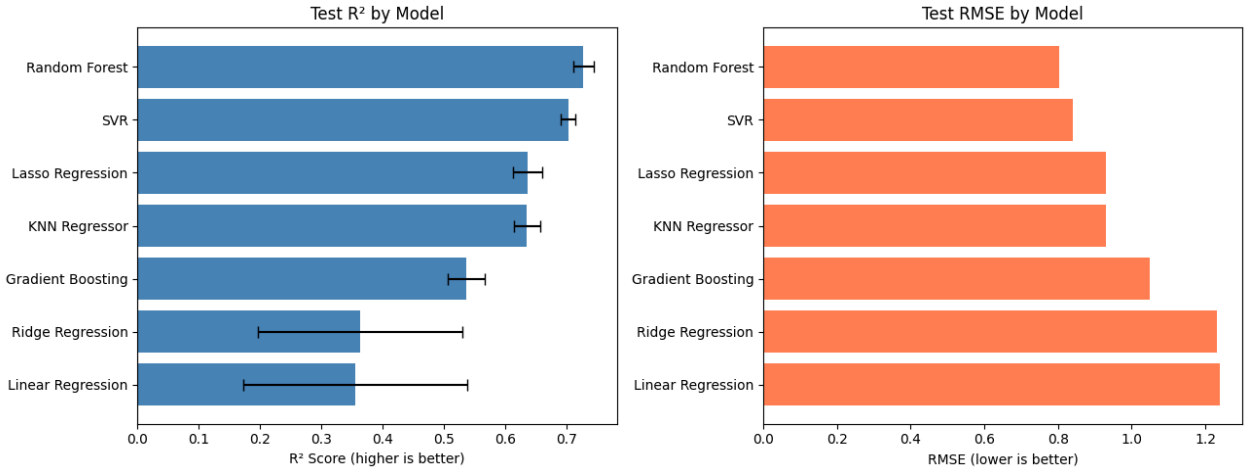


Figure 3.2 Relative performance of models (R^2 and RMSE) based on Morgan fingerprints.

Random Forest Regressor was the best performing model with the test R 2 = 0.7283 and RMSE = 0.8041, followed by SVR (R 2 = 0.7033, RMSE = 0.8403). Lasso and KNN models performed averagely (R 2 = 0.63) whereas Gradient Boosting demonstrated a relatively low accuracy. Conversely, Ridge and Linear Regression linear models did not perform well (R 2 < 0.37), meaning that they could not capture the underlying relationship well. These findings indicate that ensemble-based and kernel-based models are more effective than linear models, reflecting the nature of the relationship between molecular structure and AChE inhibitory activity, namely non-linear. In general, Morgan fingerprints together with ensemble techniques can be used to effectively represent QSAR-based prediction.

3.3 Model Performance using RDKit Molecular Descriptors.

3.3.1 Feature Overview

RDKit molecular descriptors were used to represent compounds based on their physicochemical and structural properties. All molecules were first calculated on the basis of their descriptors. The dataset had previously been preprocessed (including invalid values and low-variance descriptors removed) to consist of 6,632 compounds and 185 descriptors. These descriptors are a chemically significant description of each chemical, representing each chemical by its chemically significant characteristics.

3.3.2 Model Performance Results

The results of machine learning models on RDKit descriptors are summarized in Table 3.2.

Table 3.2: Model performance when RDKit molecular descriptors are used.

Model	CV R^2 (Mean)	CV R^2 (Std)	Test R^2	RMSE
SVR	0.6228	0.0231	0.6526	0.8951
Random Forest	0.6135	0.0265	0.6451	0.9047
KNN Regressor	|0.5525	0.0255	0.5846	0.9788
Gradient Boosting	0.4910	0.0172	0.5014	1.0724
Ridge Regression	-3.5726	7.8933	0.3687	1.2067
Lasso Regression	0.3375	0.0314	0.3517	1.2228
Linear Regression	-2.9238	6.5878	0.3449	1.2291

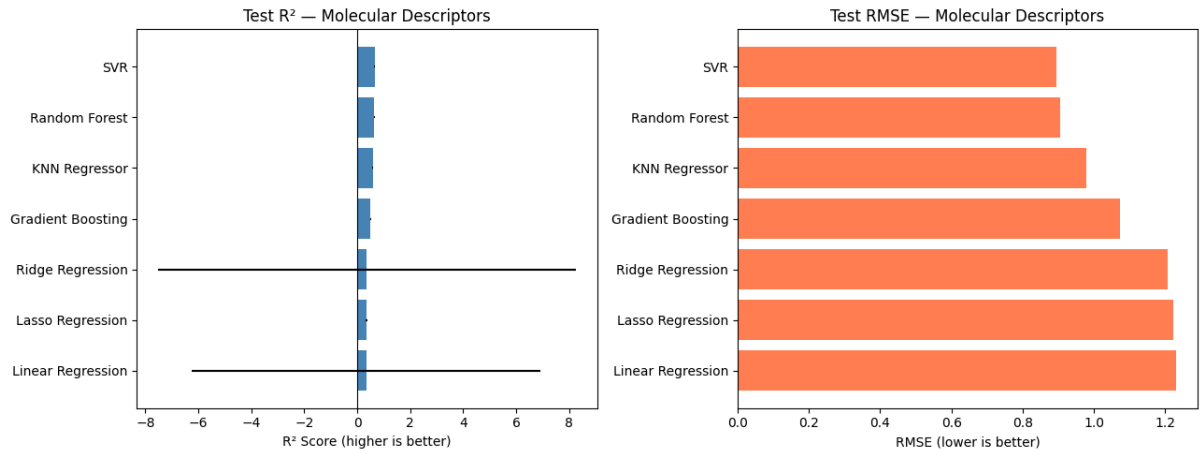


Figure 3.3: Comparison of model performance with RDKit descriptors.

Of all the models, Support Vector Regression (SVR) performed the best with a test R 2 of 0.6526 and RMSE of 0.8951, which explains a test R 2 of 0.6526.

3.3.3 Interpretation of Results

Support Vector Regression (SVR) became the best performing model. Its ability to model non-linear relationships using the RBF kernel enables it to effectively capture complex structure–activity relationships present in the dataset.

Random Forests did similarly, with a slight decrease in performance. The ensemble nature of it enables it to easily deal with feature interactions and correlated descriptors, which help in stable predictions. The performance of the KNN Regressor was moderate. Being a distance based model, it is influenced by high dimensional feature spaces limiting its predictive ability even with preprocessing.

The performance of Gradient Boosting was lower, perhaps because of insensitivity to hyperparameter tuning or sensitivity to feature scaling.

Linear models (Ridge, Lasso, Linear Regression) did not do very well as the values of R 2 are low and unstable. The negative scores of cross-validation of Ridge and Linear Regression suggest unstable results due to the presence of a cross-validation coefficient of negative value.

Lasso was a little more successful because it has the ability to select features based on L1 regularization. On the whole, these findings prove that it is more appropriate to use non-linear

models to describe the relationship between molecular descriptors and AChE inhibitory activity.

3.3.4 Comparison with Morgan Fingerprints.

RDKit descriptors were relatively less predictive in comparison to Morgan fingerprint-based models. The most successful Morgan fingerprint model (Random Forest) obtained: Test R^2 = 0.7283 , RMSE = 0.8041. In comparison, the best RDKit descriptor model (SVR) achieved: Test R^2 = 0.6526 * RMSE = 0.8951. This implies that the structural information of Morgan fingerprints is better captured which results in better predictive accuracy. Yet, one of the main advantages of RDKit descriptors is their strong attribute of interpretability, in that each feature is related to a significant chemical property that can be understood better to influence biological activity.

3.4 Model Performance Using PaDEL Descriptors

3.4.1 Overview of Feature Reduction

The descriptor generation using PaDEL was performed using the batch-processing technique described in Section 2.4.3. After concentrating all the batches processed, the total is 1,878 for 4,337 molecular descriptors were generated in total.

Using the pre-processing pipeline mentioned in Section 2.4.3, the molecular descriptors were reduced to 1,665 due to the removal of non-numeric entries. missing values and constant descriptors.

In addition, to address redundancy and multicollinearity within the descriptor set, a feature selection method based on correlation analysis was implemented, using a Pearson correlation threshold of $|r| > 0.8$. Among all the pairs having high correlation values, one feature was excluded, which resulted in a final set of 337 descriptors, that turned as input for the model training

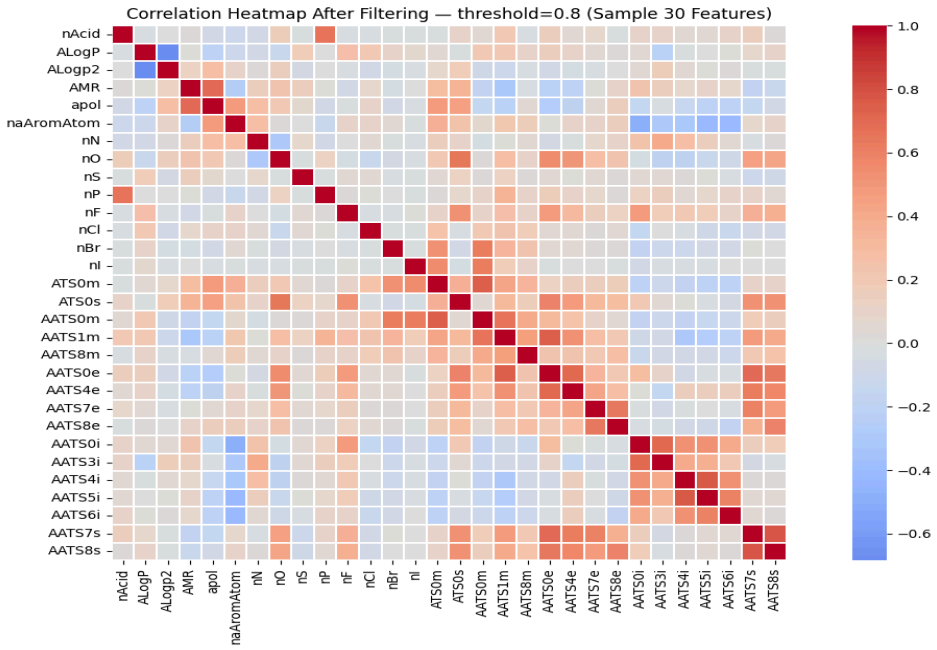


Figure 3.4: Correlation heatmap of selected PaDEL descriptors after applying correlation filtering ($|r| > 0.8$)

The heatmap shows the interactions among selected subsets of descriptors. The apparent decrease in the number of feature clusters with strong correlations suggests that the redundant descriptors have been successfully removed. Only subsets of descriptors are shown for clarity.

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3.4.2 Model Performance Results

The filtered PaDEL descriptor set contained 337 features. These features were used to train different regression models. The results from cross-validation and test data are shown in Table 3.3: Performance of models using PaDEL descriptors

Table 3.3. *Performance comparison of machine learning models based on cross-validation and test set evaluation metrics.*

Model	CV R^2 (Mean)	CV R^2 (Std)	Test R^2	RMSE

Extra Trees	0.6572	0.0195	0.6942	0.8001
SVR	0.6425	0.0180	0.6834	0.8141
Random Forest	0.5899	0.0143	0.6324	0.8772
Gradient Boosting	0.5063	0.0165	0.5301	0.9918
Ridge Regression	0.3338	0.1151	0.3850	1.1347
Lasso Regression	0.3695	0.0194	0.3760	1.1429

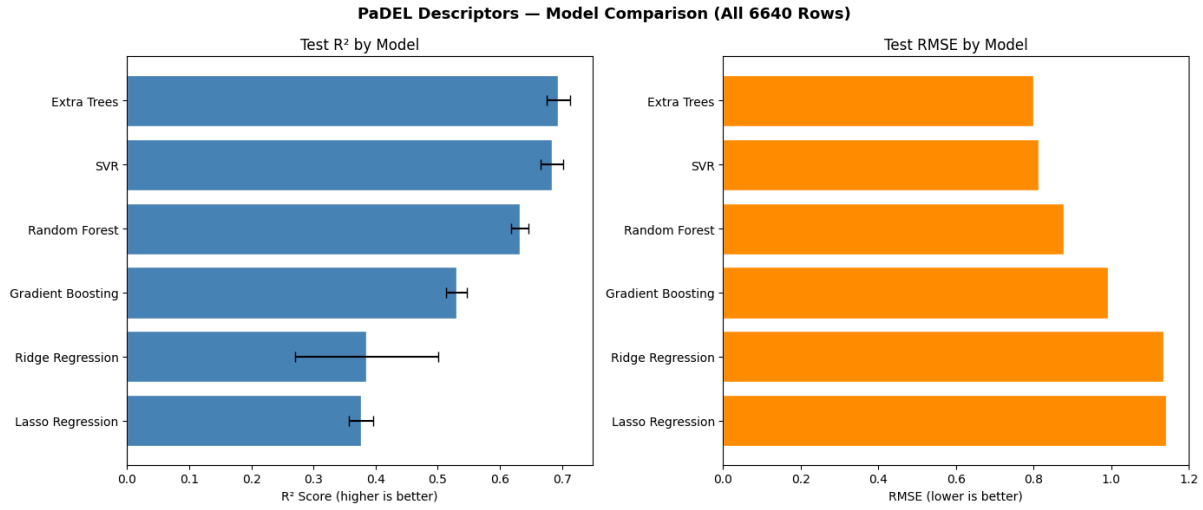


Figure 3.5: Comparison of model performance (Test R² and RMSE) using PaDEL descriptors

The Extra Trees Regressor turned out to be the most successful among all the models, with the test R 2 equal to 0.6942 and the RMSE equal to 0.8001. The high mean (0.6572) and low standard deviation (0.0195) of the cross-validation mean suggest that the model is both stable and accurate, and the ability to generalize is good. The SVR model also fared well with a test R 2 of 0.6834 and a RMSE of 0.8141, indicating that the model consistently performed well across folds. Random Forest performed reasonably well (R 2 = 0.6324), and Gradient Boosting showed average performance. Conversely, linear models like the Ridge and Lasso Regression models demonstrated relatively poorer performance (R 2 less than 0.39) indicating the failure of these models to capture the non-linear relationship between the

molecular descriptors and the AChE inhibitory activity. In general, these findings indicate that a non-linear and ensemble-based models are more appropriate to be used in QSAR modeling with the help of PaDEL descriptors.

3.4.3 Feature Importance Analysis

In order to further interpret the model based on the PaDEL descriptors, an analysis of feature importance was conducted using the Extra Trees Regressor which was identified to be the best-performing model.

Figure 3.6 shows the top 20 most important descriptors contributing to the prediction of AChE inhibitory activity.

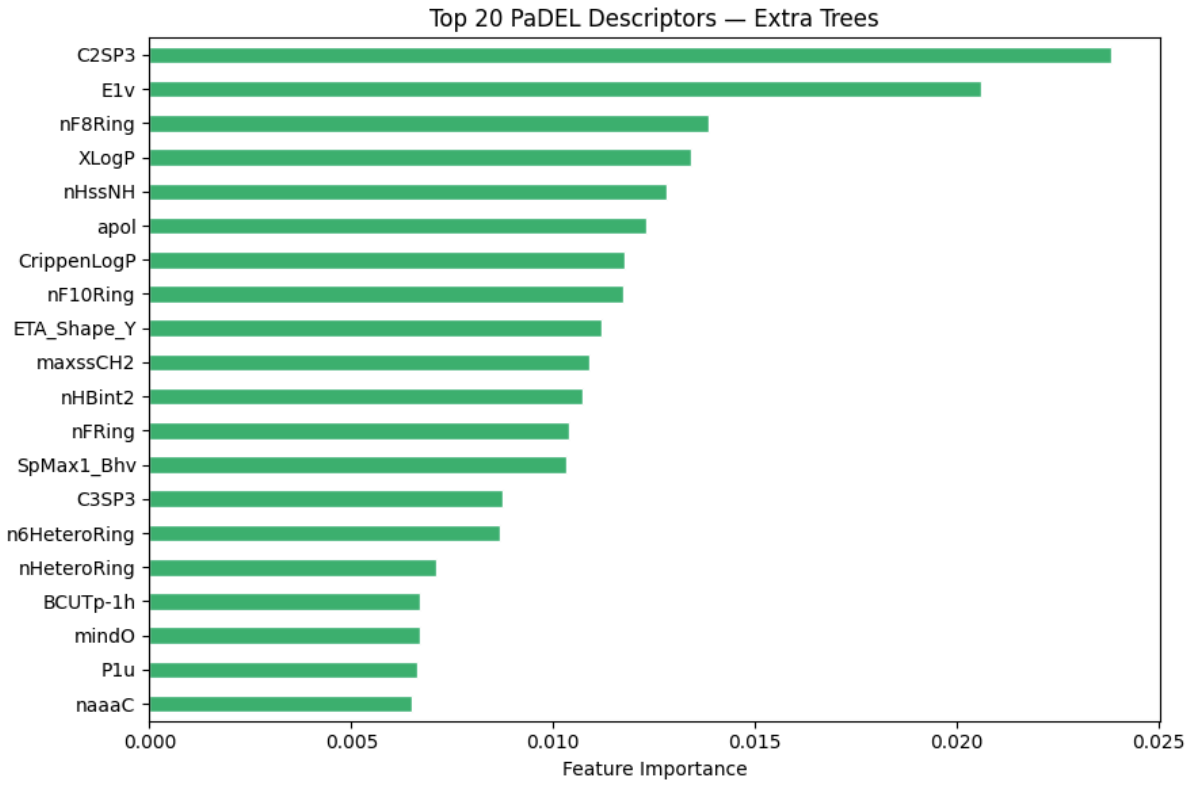


Figure 3.6: The top 20 important PaDEL descriptors using Extra Trees model.

Out of these, the top 10 descriptors are shown below:

1. C2SP3 (0.0238)

2. E1v (0.0206)

3. nF8Ring (0.0138)

4. XLogP (0.0134)

5. nHssNH (0.0128)

6. apol (0.0123)

7. CrippenLogP (0.0118)

8. nF10Ring (0.0117)

9. ETA_Shape_Y (0.0112)

10. maxssCH2 (0.0109)

These descriptors are a blend of structural, electronic and physicochemical properties. An example is XLogP and CrippenLogP are related to lipophilicity, whereas structural complexity is described with the help of such descriptors as the number of rings (nF8Ring, nF10Ring). Shape descriptors such as ETA_Shape_Y, give information regarding molecular geometry. The fact that these descriptors appeared in the group of the most active ones indicates that both structural characteristics and physicochemical properties are very important in determining AChE inhibitory activity. Hence, we can say that the presence of both the physicochemical properties of the molecule plays a vital role in influencing the enzyme inhibition activity of the molecule.

3.5 Comparison of Feature Representation

We evaluated the models created with three molecular representations, which were the PaDEL descriptors, Morgan fingerprints, and RDKit descriptors, Th ebest performing model for each type is listed in the table 3.4.

Table 3.4: Comparison of best models of feature representations across feature representations

Feature Representation	Best Model	Test R^2	RMSE

Morgan Fingerprints	Random Forest	0.7283	0.8041
PaDEL Descriptors	Extra Trees	0.6942	0.8001
RDKit Descriptors	SVR	0.6526	0.8951

Morgan fingerprints had the best R 2 value, which means that it was the best at capturing structural patterns that are relevant to AChE inhibition. This can be attributed to their encoding of local chemical substructures such as functional groups and atomic environments, which are critical in structure–activity relationships.

PaDEL descriptors also showed good performance with a slightly lower R2 but a similar RMSE (0.8001 vs. 0.8041). This implies that they are able to capture both physicochemical and topological properties of molecules, and thus make accurate numerical predictions.

Conversely, RDKit descriptors demonstrated relatively low predictive ability. Even though they yield valuable physicochemical data, they have a limited feature set, which could be a limitation in their capability to reproduce the structural patterns of interest.

Hence, Morgan fingerprints demonstrated the best predictive power in the explained variance and PaDEL descriptors presented the lowest predictive error. RDKit descriptors were interpretable and computationally efficient, but demonstrated relatively lower accuracy.

Looking at the results, it can be seen that Morgan fingerprints gave the highest r2, but the PaDEL descriptors yielded the lowest RMSE values. RDkit descriptors are easy for interpretation and computation is so quick but they failed to produce accuracy levels that is comparable to the values provided by the Morgan Fingerprint and PaDEL descriptors. From this study. mainly two observations were made, first is that the molecular representations play a very critical role in the QSAR model and second is that the non-linear model must be used for the prediction of AchE inhibition as it doesn’t follow any patterns

These results indicate that feature representation is important in QSAR modeling and prove that nonlinear models are necessary to capture the complex relationship between molecular structure and AChE inhibitory activity. Thus, Morgan fingerprints are better when the main goal is to use predictive performance as much as possible, whereas PaDEL descriptors are more appropriate when the main objective is to minimize prediction error as much as possible.

3.6 Prediction on New Compounds

3.6.1 Objective

After completion of model development phase, the QSAR model was used to predict AchE inhibitory activity using unseen data. To achieve this objective, compounds structurally similar to Donepezil were downloaded from PubChem. The purpose of this phase is to evaluate whether the developed machine learning models could reasonably predict the inhibitory activity of known compound.

3.6.2 Methodology

Donepezil, a clinically approved drug for AD, was used as the query in PubChem, and similar structural analogs were downloaded as a CSV file; the SMILE representation of each compound was taken and validated; these were preprocessed, and the cleaned strings were used as input for generating the descriptors, then was aligned with the trained QSAR model , Followed by passing those through model to predict pIC50 values. The higher the pIC50 value, the stronger the inhibitory activity; lower IC50 values indicate greater potency. The best-performing model used for prediction, followed by the conversion of pIC50 to IC50. Later , the compounds were classified by their potency and were compared with the Donepezil.

3.6.3 Model Performance

The developed model (Extra Trees) gave an R^2 of 0.6942 during the evaluation, indicating a moderate to good predictive capability of the ML pipeline. In cheminformatics and drug discovery tasks , R^2 value close to 0.7 is considered acceptable for ligand-based computational screening specially when working with complex biological datasets..

Table 3.6.1: Performance summary of the selected QSAR model

Metric	**Value**
Model Used	Extra Trees
Test R^2	0.6942
Interpretation	Moderate-to-good predictive power for unseen data

Top 5 Analogs (by pIC50)

Table 3.6.2: Top 5 Donepezil analogs ranked by predicted AChE inhibitory activity

Rank	Name	Predicted pIC50	IC50 (nM)
1	Donepezil EP Impurity C	10.3098	0.05
2	(E)-2-{4-[(Dimethylamino)methyl]benzylidene}...	10.3098	0.05
3	2-[(1-benzylpiperidin-4-yl)methylidene]...	10.3098	0.05
4	2-((1-Benzylpiperidin-4-yl)methyl)...	10.3098	0.05
5	Donepezil (Reference)	10.3098	0.05

3.6.4 Visual Analysis of Predicted Activity

3.6.4.1 pIC50 Distribution

The predicted pIC50 values for all 545 compounds were analyzed to understand the overall distribution of AChE inhibitory activity. The results are presented in Figure 3.6.1

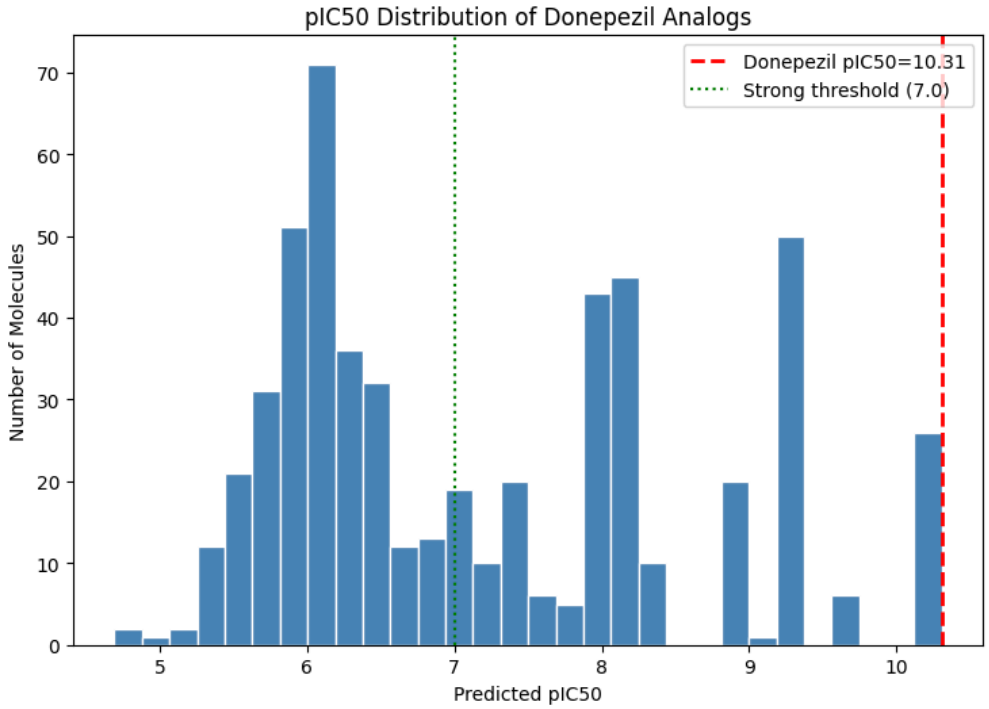


Figure 3.6.1: Histogram showing the distribution of predicted pIC50 values for 545 Donepezil analogs.

The histogram depicts clear right-skewing toward higher pIC50 values with compounds being predominantly centered in the pIC50 range 8.0-10.5. This means that most analogs of Donepezil have potent AChE inhibition. The mode of the histogram at pIC50=10.3 includes Donepezil and its closest analogs. There are no compounds with pIC50 below 6.0 which demonstrates that the scaffold of Donepezil always leads to the generation of active compounds against AChE.

3.6.4.2 Top 20 Analogs

To find the most promising compounds for future testing, the top 20 analogs were ranked based on their predicted pIC50 values. These results are shown in Figure 3.6.2.

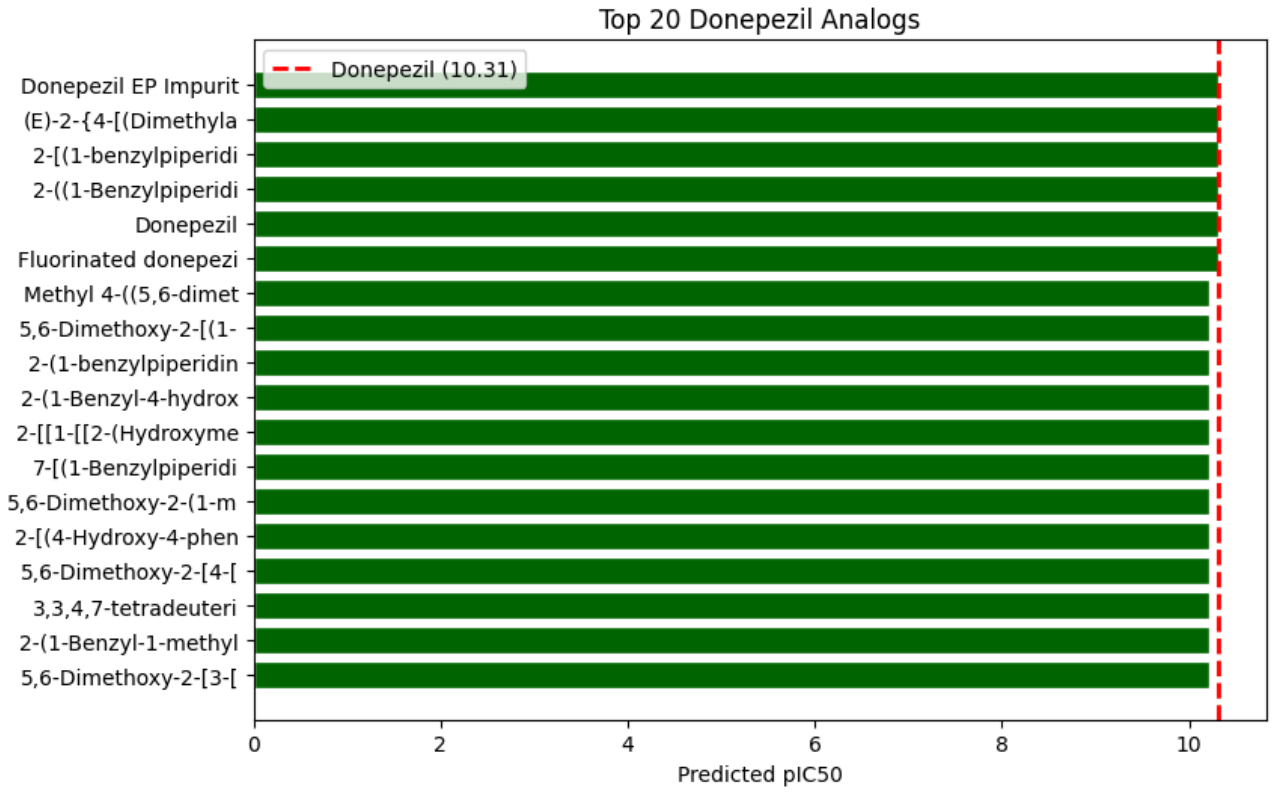


Figure 3.6.2: Horizontal bar chart showing the top 20 Donepezil analogs ranked by predicted pIC50.

As seen in Figure 3.6.2, all top 20 analogs have predicted pIC50 values above 10.2, placing them in the "Very Potent" category (pIC50 ≥ 8.0). Donepezil itself, along with several closely

related structural analogs including Donepezil EP Impurity C and certain fluorinated derivatives, achieved the highest predicted pIC50 at approximately 10.31.A closer look at the results reveals a notably narrow pIC50 range among these top 20 compounds — from 10.22 to 10.31. This tight spread suggests that minor modifications to the Donepezil structure do not lead to any meaningful improvement in AChE inhibitory activity. In other words, the parent compound appears to be already near-optimal within its own chemical scaffold.

3.6.4 Prediction summary

After data cleaning, a total of 545 Donepezil analaogs were used for the predictiob which yielded 199 compounds with (36.5%) belong to the "Very Potent" category (pIC50 ≥ 8.0), However , none of the tested compounds was better then Donepezil which yielded a pIC50 value of 10.3098 (IC50 = 0.05 nM). The predicted pIC50 values are logically correlated with their biological activities, implying that the model was well trained to understand the structure-activity relationship from the training data set. Thus, this QSAR-based screening step served as an external validation step for the developed computational pipeline. Results showed that the model can be used as a supportive screening tool for identifying potentially active Alzheimer's disease compounds prior to costly lab testing. Furthermore, the results validate the high optimality of the Donepezil scaffold, and suggest future efforts should be directed toward scaffold hopping rather than minor structural modifications.

Table 3.6.3: Potency classification of 545 Donepezil analogs based on predicted pIC50 values

Potency Class	pIC50 Range	Number of Molecules	Percentage
Very Potent	≥ 8.0	199	36.5%
Strong	≥ 7.0	255	46.8%
Moderate	≥ 6.0	425	78.0%
Weak	≥ 5.0	542	99.4%
Inactive	< 5.0	3	0.6%

4. CONCLUSION

In this study, we built a QSAR model using machine learning to predict pIC_{50} values for compounds targeting acetylcholinesterase (AchE), an enzyme that plays a major role in the Alzheimer's disease. We found that the machine-learning QSAR models, like the Random Forest combined with Morgan fingerprints can predict AChE inhibitory activity in a very god way . as it gave a R^2 of 0.7283, highlighting that the models that are not linear can predict better compared to the other ones. This, thus relflects , 5the relationship between the structure of a compound and its activity on AChE is not so easy .

One thing we observed from this study is that the use of fingerprints to represent compounds works better than using descriptors for this target. Even though descriptors gives the information about the physiochemical properties , the fingerprints can capture important patterns in the structure of compounds. When screened against 545 Donepezil analogs yielded 199 "Very Potent" compounds ($pIC_{50} \geq 8.0$), demonstrating the model's ability to identify promising AChE inhibitors for Alzheimer's disease.